

Bernardo Micol Righi

Back-End Developer · AI Systems · Tech Influencer

+55 (51) 99601-1501

bernardomicolrighi@outlook.com

righi.dev

github.com/righibe

linkedin.com/in/bernardo-righi

lattes.cnpq.br/8608406696939587

PROFESSIONAL SUMMARY

Back-end developer with 2 years of experience in Python, AI, and automation. Researcher at Unisinos in partnership with Dell, focused on LLM agent observability using LangChain and OpenTelemetry. Co-founded the [Servidor dos Programadores](#), one of the top-6 dev communities in Brazil on Discord (18k+ members). I also create content about programming and AI as a tech influencer. Fluent in English.

PROFESSIONAL EXPERIENCE

Researcher — AI Agents & Observability

[Unisinos / Dell Technologies](#)

Jun 2026 – Present · Remote

- Researching observability patterns for multi-agent systems, studying how to trace behavior, latency, and failures of AI agents in production.
- Investigating strategies to trace tool calls, agent handoffs, and context propagation — project in a research and architecture-prototyping phase alongside the Dell team.
- Delivered a comparative study of 6 embedding-clustering pipelines (K-Means, UMAP, PaCMAP, LocalMAP, densMAP + HDBSCAN) on real support-ticket data, with statistical validation (trustworthiness, DBCV) grounded in the scientific literature, informing the team's approach to automatic topic discovery.

Co-founder & Back-End Developer

[Servidor dos Programadores](#)

Jun 2022 – Present · Remote

- The community was growing fast but losing members due to poor onboarding and moderation — I built Python/Discord.py bots that automated welcomes and technical support, improving retention and reducing the team's manual work.
- As growth accelerated, the infrastructure needed zero-downtime deploys — I set up CI/CD pipelines with GitHub Actions that automatically test and ship updates on every push, eliminating manual deploys.
- Led the organization of technical events with sponsors like Alura, Shard Cloud, and Hostinger, coordinating speakers and live sessions for 18k+ developers.

TECHNICAL SKILLS

Languages: Python Java C C++ JavaScript SQL

AI & Agents: LangChain LangGraph LCEL RAG LLMs scikit-learn

Observability: OpenTelemetry Jaeger Grafana Agent Tracing

DevOps & Infra: Docker GitHub Actions CI/CD Git Linux

Frameworks: Spring Boot Django Flask Discord.py

Databases: MySQL JPA / Hibernate

EDUCATION

B.Sc. in Computer Science

Universidade do Vale do Rio dos Sinos (Unisinos)

Feb 2026 – Present · Sao Leopoldo, Brazil · [lattes.cnpq.br/8608406696939587](#)

COURSES & CERTIFICATIONS

Java Marathon (Java Virado no Jiraya) · Python Worlds 1, 2 & 3 · HTML5 & CSS3 (Curso em Video) · DevOps & Python Immersion (Alura) · Design and Social Enterprise HDSE Leadership (Lehigh University)

LANGUAGES

Portuguese — Native | **English** — Fluent (C1) | **Spanish** — Intermediate (B1)

PROJECTS

AI Technical Support Agent

Python · LLMs · OpenAI GPT-4o · RESTful APIs  github.com/righibe/technical-suport-Agent

- Situation: first-line technical support tickets needed faster triage without waiting on a human agent.
- Action: built an autonomous agent in Python powered by GPT-4o to interpret tickets and generate contextual resolutions through RESTful APIs.
- Result: automated first-level responses for common cases without human escalation.

CI/CD Pipeline in Python

Python · GitHub Actions · Docker · Flask  github.com/righibe/ci-cd-pipeline-python

- Situation: no automated testing or deployment process — every update was a production-bug risk.
- Action: built a complete pipeline with GitHub Actions and Docker, running automated tests on every push and deploying to the container only if everything passed.
- Result: eliminated manual deploys and gained the guarantee that only validated code reached the environment.

Generative AI Discord Bot

Python · Discord.py · Open-source LLM API  github.com/righibe/bot-discord-IAgenerativa

- Situation: the community wanted real-time AI assistance inside Discord without relying on paid APIs.
- Action: built a bot with Discord.py integrating an open-source generative AI API, handling multi-turn conversation context.
- Result: gave 18k+ community members free, real-time AI support directly inside the server.

Housing Price Prediction — Sao Paulo

Python · scikit-learn · Decision Tree · Random Forest  github.com/righibe/predict-prices-sp

- Situation: wanted to apply supervised ML to a real problem, moving beyond theory.
- Action: trained and compared Decision Tree and Random Forest with scikit-learn to predict housing prices in Sao Paulo, analyzing feature importance.
- Result: identified location and square footage as the dominant variables and solidified my hands-on foundation in regression.

CRUD REST API — Django & SQL

Python · Django · REST Framework · MySQL  github.com/righibe/CRUD-Python-Django

- Situation: needed hands-on practice with Django's ORM and REST patterns beyond Flask.
- Action: built a full CRUD REST API with Django connected to a SQL database, structuring validation and serialization.
- Result: solidified Django fundamentals and REST API design patterns transferable to production work.

People Registry REST API — Java

Java · Spring Boot · JPA / Hibernate · Spring Web  github.com/righibe/Cadastro-pessoas

- Situation: only had Python experience and needed to understand the Java ecosystem.
- Action: built a REST API from scratch with Spring Boot and JPA, following the Controller → Service → Repository flow with DTO and ORM.
- Result: a solid foundation in Java and clarity on the MVC pattern for enterprise projects.